

Auteur: Wout Vanderborght en Tyas Hermans

Studierichting: Informatica

Oefeningenreeks 7C nr. 2.2

$$f : \mathbb{R} \rightarrow \mathbb{R} : (x, y) \rightarrow \sqrt{4x^2 + y^2}$$

$$\frac{\partial f}{\partial x}(x, y) = \frac{4x}{\sqrt{4x^2 + y^2}}$$

$$\frac{\partial f}{\partial y}(x, y) = \frac{y}{\sqrt{4x^2 + y^2}}$$

$$\frac{\partial^2 f}{\partial x^2}(x, y) = \frac{4\sqrt{4x^2 + y^2} - \frac{16x^2}{\sqrt{4x^2 + y^2}}}{4x^2 + y^2} = \frac{4(4x^2 + y^2) - 16x^2}{(4x^2 + y^2)\sqrt{4x^2 + y^2}} = \frac{4y^2}{(4x^2 + y^2)\sqrt{4x^2 + y^2}}$$

$$\frac{\partial^2 f}{\partial xy}(x, y) = \frac{-4xy}{(4x^2 + y^2)\sqrt{4x^2 + y^2}}$$

$$\frac{\partial^2 f}{\partial yx}(x, y) = \frac{-4xy}{(4x^2 + y^2)\sqrt{4x^2 + y^2}}$$

$$\frac{\partial^2 f}{\partial y^2}(x, y) = \frac{\sqrt{4x^2 + y^2} - \frac{y^2}{\sqrt{4x^2 + y^2}}}{4x^2 + y^2} = \frac{4x^2}{(4x^2 + y^2)\sqrt{4x^2 + y^2}}$$

$$D^2 f = \begin{pmatrix} \frac{4y^2}{(4x^2 + y^2)\sqrt{4x^2 + y^2}} & \frac{-4xy}{(4x^2 + y^2)\sqrt{4x^2 + y^2}} \\ \frac{-4xy}{(4x^2 + y^2)\sqrt{4x^2 + y^2}} & \frac{4x^2}{(4x^2 + y^2)\sqrt{4x^2 + y^2}} \end{pmatrix}$$

References